CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE)

April 2013

CE214 MICROPROCESSORS & INTERFACES

Date: 27.04.2013, Saturday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions. **[11]**

[A] What is difference between STA and STAX instructions? [1]

[B] What is an assembler? [1]

[C] How does the microprocessor differentiate among a positive number, a negative number and a bit pattern? [1]

[D] When does microprocessor check for interrupt? [1]

(i) after completion of program execution
(ii) at start of program execution
(iii) at the end of execution of current instruction
(iv) at the end of execution of current T-state of current instruction

[E] What is functionality of RIM instruction? [1]

[F] Write down the addressing modes of following instructions [3]

(i) MOV Rd, Rs
(ii) ADD R
(iii) ADI 8-bit
(iv) IN 8-bit port address
(v) JM 16-bit address
(vi) ANI 8-bit

[G] List and write description of various conditional Jumps. [3]

Q.2 Answer the following questions. **[12]**

[A] By taking suitable sequence of instruction, write down how to initialize Programmable Interrupt Controller. [5]

[B] What is purpose of DMA? How is it interfaced with 8085? What are the internal components of DMA? [7]

OR

[B] A Set of ten bytes are stored in memory starting with the address x2050H. Write a program to check each byte, and save the bytes that are higher than 60_{10} and lower than 100_{10} in memory location starting from x2080H. [7]
Data Bytes: 49,5A,4B,5C,3A,A3,A5,10,28,FA

Q-3 Answer the following questions.(Attempt any Two) **[12]**

[A] Write an assembly program to add the two hexadecimal numbers 3AH and 48H and to display the answer at an output port.

[B] Explain the functional diagram of 8155 and explain its control word.

[C] 8085 has five types of interrupt. Write down characteristic of each type of interrupt.

SECTION-II

Q-4 Answer the following questions.

[A] After RESET 8086 fetches its first instruction from ________ address. [1]

[B] Why devices which are interfaced on system bus must have tri-state logic? [1]

[C] Which signal is used to interface slower devices with 8085? [1]

[D] Why is the need to demultiplex the bus AD7-AD0? [1]

[E] Which signal is used to interface the devices which send data at a high speed to 8085? [1]

[F] Assume that content of memory location 2075H is 47H. During the execution of LDA 2300H instruction, the 2[nd] and 3[rd] T-sates of 4[th] machine cycle, Specify the contents of the address bus A15-A8 and the data bus AD7-AD0. [3]

[G] 2100 LXI SP, 3000H [3]
2103 CALL 2300H
After the execution of CALL instruction, specify in which memory locations return address will be stored and what will the content of SP?

Q-5 Answer the following questions. **[12]**

[A] State addressing mode for the following instructions : [5]
1) CPI 55H 2) STA 3200H 3) ADD M 4) MOV A,B 5) CMA

[B] Loop: LXI B,3FFFH [4]
DCX B
JNZ Loop
When content of B and C reaches to zero, program should come out from the loop. Find out the mistake in the program and suggest the modification.

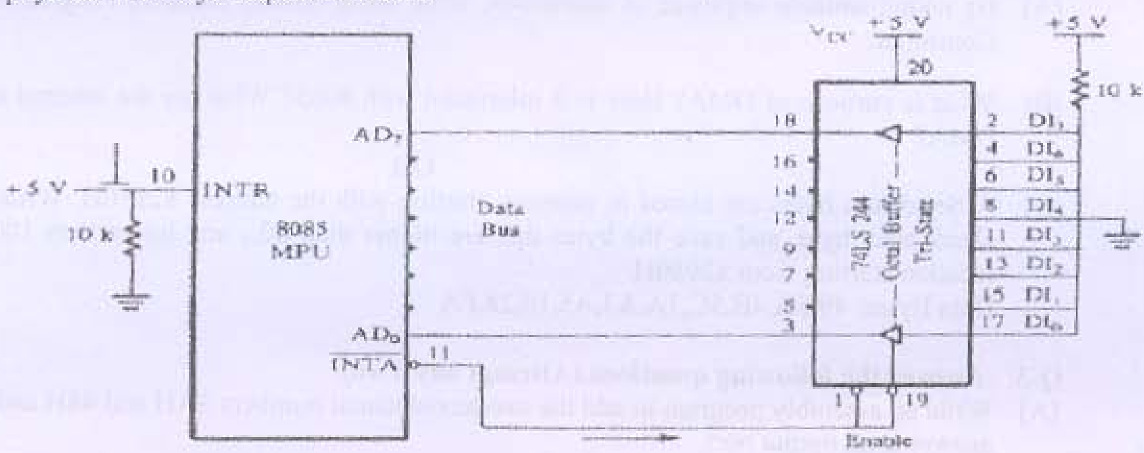
[C] Draw the circuit and Explain the principle of scanned multiplexed display to interface seven segment displays with microprocessor. State the advantage and disadvantage of this technique. [3]

OR

[A] 8085 system requires following memory map in the system : [5]
EPROM : 16 Kbytes at last 16K processor address space
8 Kbytes EPROM devices is available. Design for absolute
Decoding (partial decoding not allowed). Draw the neat circuit diagram.

[B] How 64 Terabytes of virtual memory is managed eventhough 80386 has only 4Gbytes of physical memory? Explain in detail. [4]

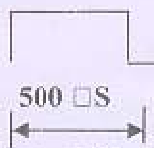
[C] [3]



Write the response of 8085 (assuming interrupt system is enabled) in detail when key is pressed.

Q-6 Answer the following questions.(Attempt any Two) **[12]**

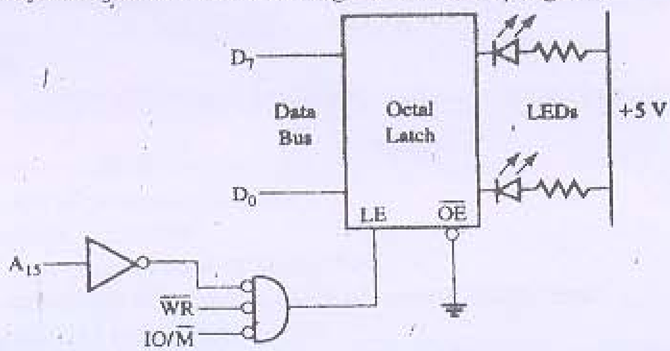
[A]



The above pulse is connected to RST6.5 and RST5.5 interrupt pins of 8085. Describe the response of 8085 for the following case if interrupt system is already enabled for RST6.5 and RST5.5 :

RST6.5 service routine takes 300 Microsecond to complete and enables the interrupt system just before returning to mail line program. RST5.5 interrupt service routine takes 150 microseconds and enables interrupt system just before returning to main line program.

[B]



Identify the port address of the output port in the above figure. Write a program to glow each LED one by one continuously.

[C] Specify memory map of the following diagram and explain significance of the don't care address line on memory addresses.